

Retail Sales Forecasting & Time series Analysis

Using Python, ARIMA, Seasonal Decomposition & Data Visualization

Dataset Preparation → *Time Series Analysis* → *Forecasting* → *Business Insights*

TOOLS & TECHNOLOGIES :

PYTHON | PANDAS | NUMPY | MATPLOTLIB | STATSMODELS | POWER BI



Project Overview

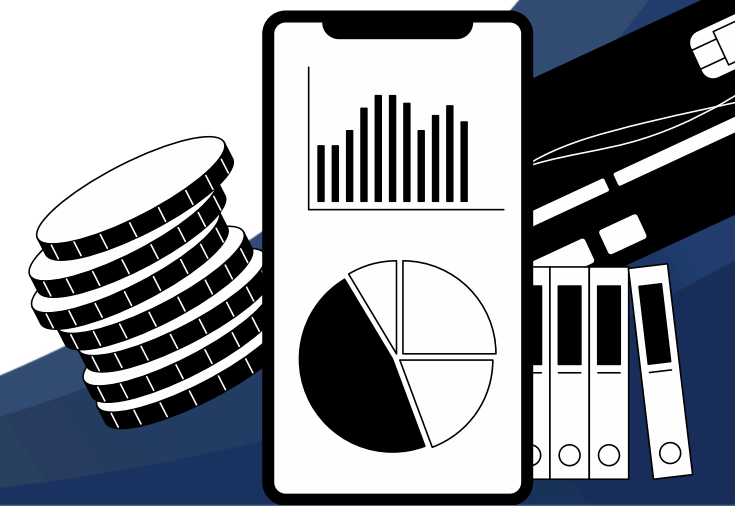
This project analyzes historical retail sales data to identify sales trends, seasonal patterns, and future demand using Time Series Analysis and Forecasting techniques. Python was used for data preparation, exploratory analysis, and ARIMA forecasting, while Power BI was utilized to create interactive dashboards for business decision-making.

Objectives:

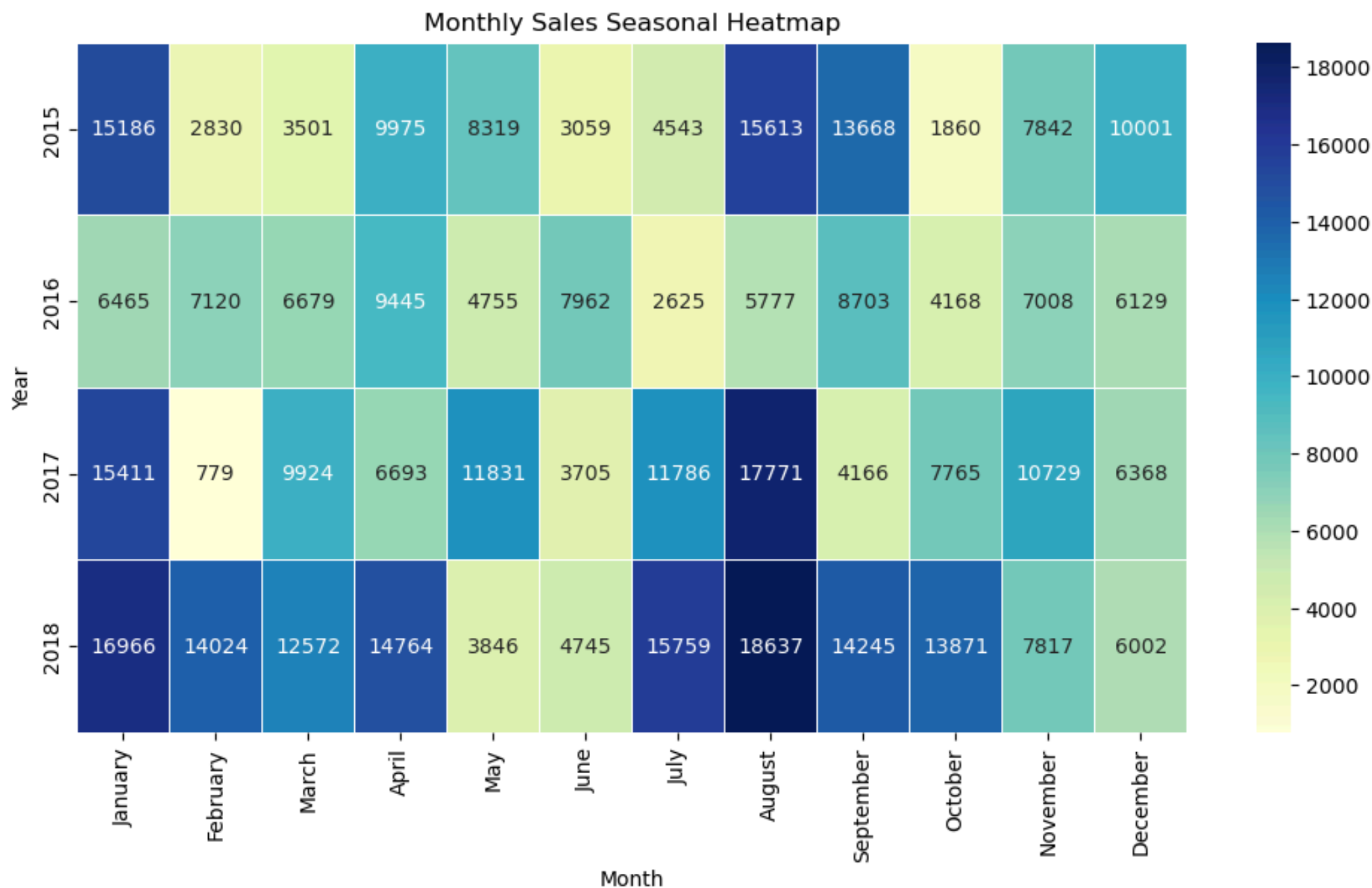
- Analyze historical sales performance
- Identify trend and seasonal demand patterns
- Forecast future sales using ARIMA
- Support inventory and sales planning decisions
- Deliver actionable business insights through dashboards

WORKFLOW

Dataset Preparation → Time Series Analysis → Forecasting → Business Insights



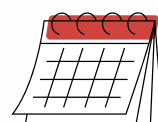
Seasonal Sales Patterns



Key Insight:



**Peak Sales month :
August 2018**



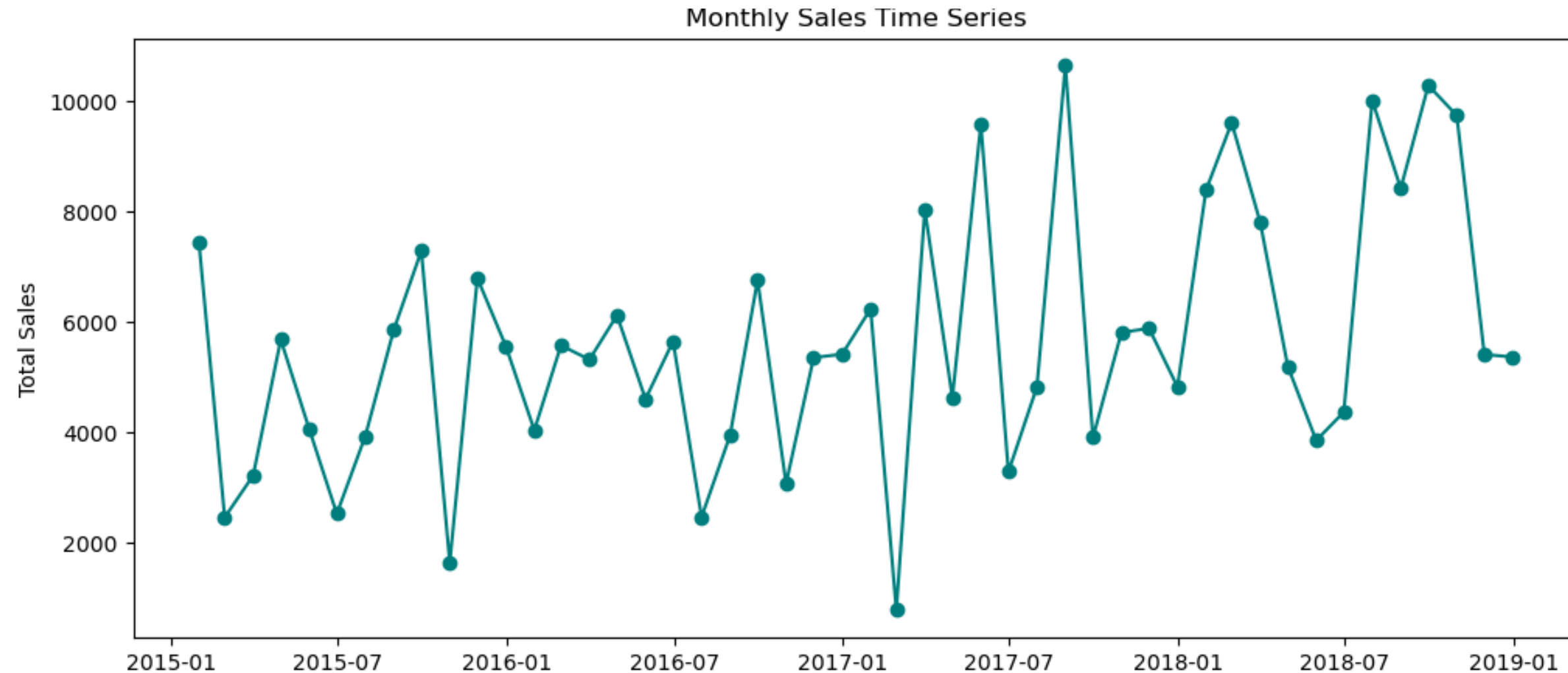
**Lowest Sales Month:
February 2017**

- Strong sales concentration observed during mid-year periods.
- Seasonal demand patterns can support inventory and resource planning.
- Certain months consistently outperformed others across years.
- Heatmap analysis helps identify high-demand business period

Business Insight:

- Increase inventory and marketing activities during peak-demand months
- Use seasonal insights for demand forecasting and sales planning
- Focus promotional strategies during lower-performing periods stabilize sales performance

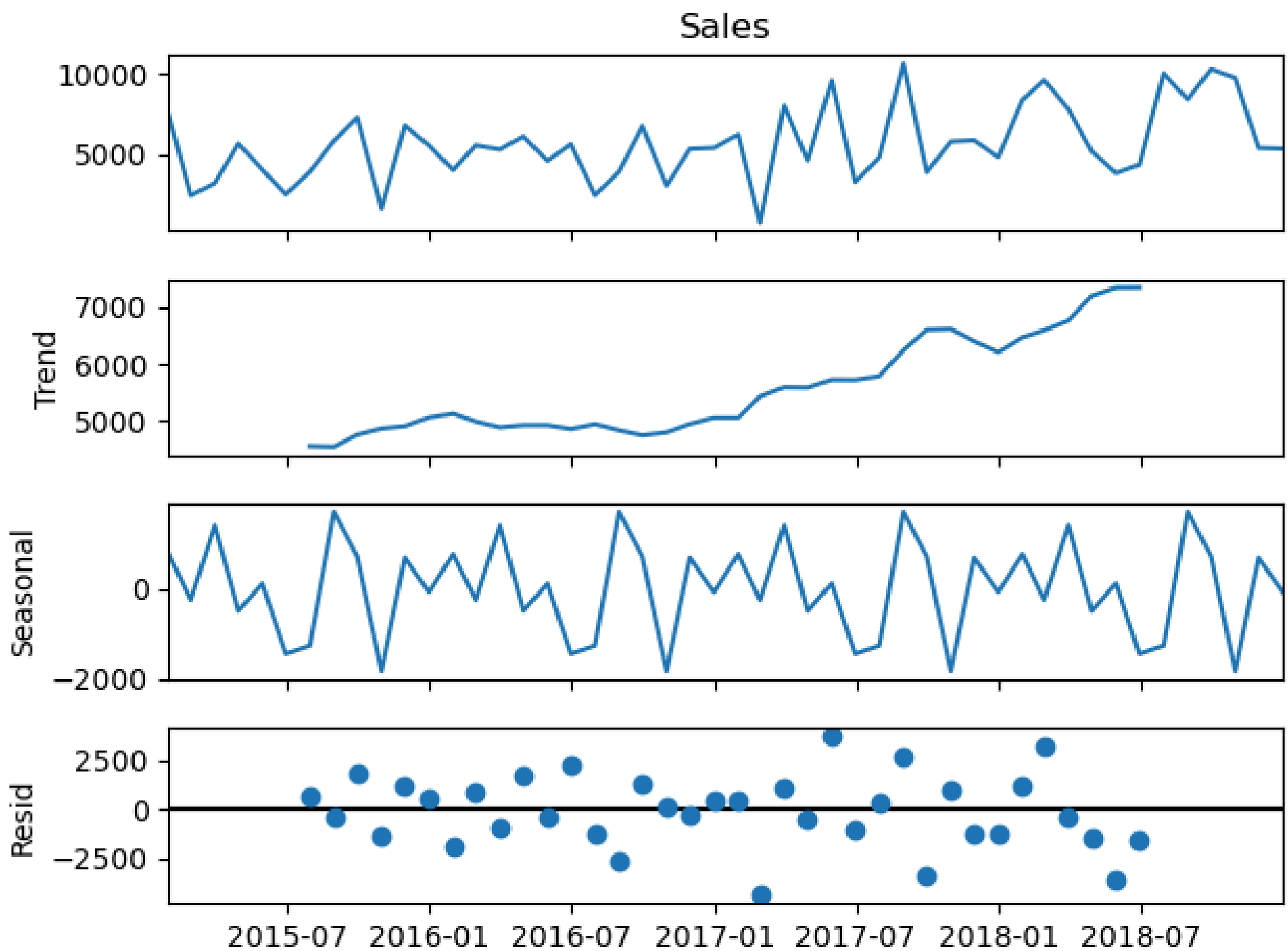
Monthly Sales Trend Analysis



Key Insight:

- Sales show an overall upward trend from 2015–2018, indicating business growth and increasing customer demand.
- Monthly sales fluctuations suggest high volatility and seasonal purchasing behavior.
- Peak sales periods exceeded 10,000, likely driven by promotions, holidays, or strong market demand.
- Sudden sales drops may indicate off-season effects or operational challenges.
- Repeating sales patterns reveal seasonality, making the data suitable for forecasting and demand prediction.
- Higher peaks in later years suggest expanding market reach and improved business performance.

Sales Decomposition Analysis



Key Insights :

Sales decomposition analysis was performed to examine the underlying trend, seasonal behavior, and irregular variations in retail sales data.

Trend Component : Shows a gradual increase in sales over time, indicating overall business growth.

Seasonal Component : Regular fluctuations suggest recurring seasonal sales patterns across different periods.

Residual Component : Residual values represent irregular variations not explained by trend or seasonality.

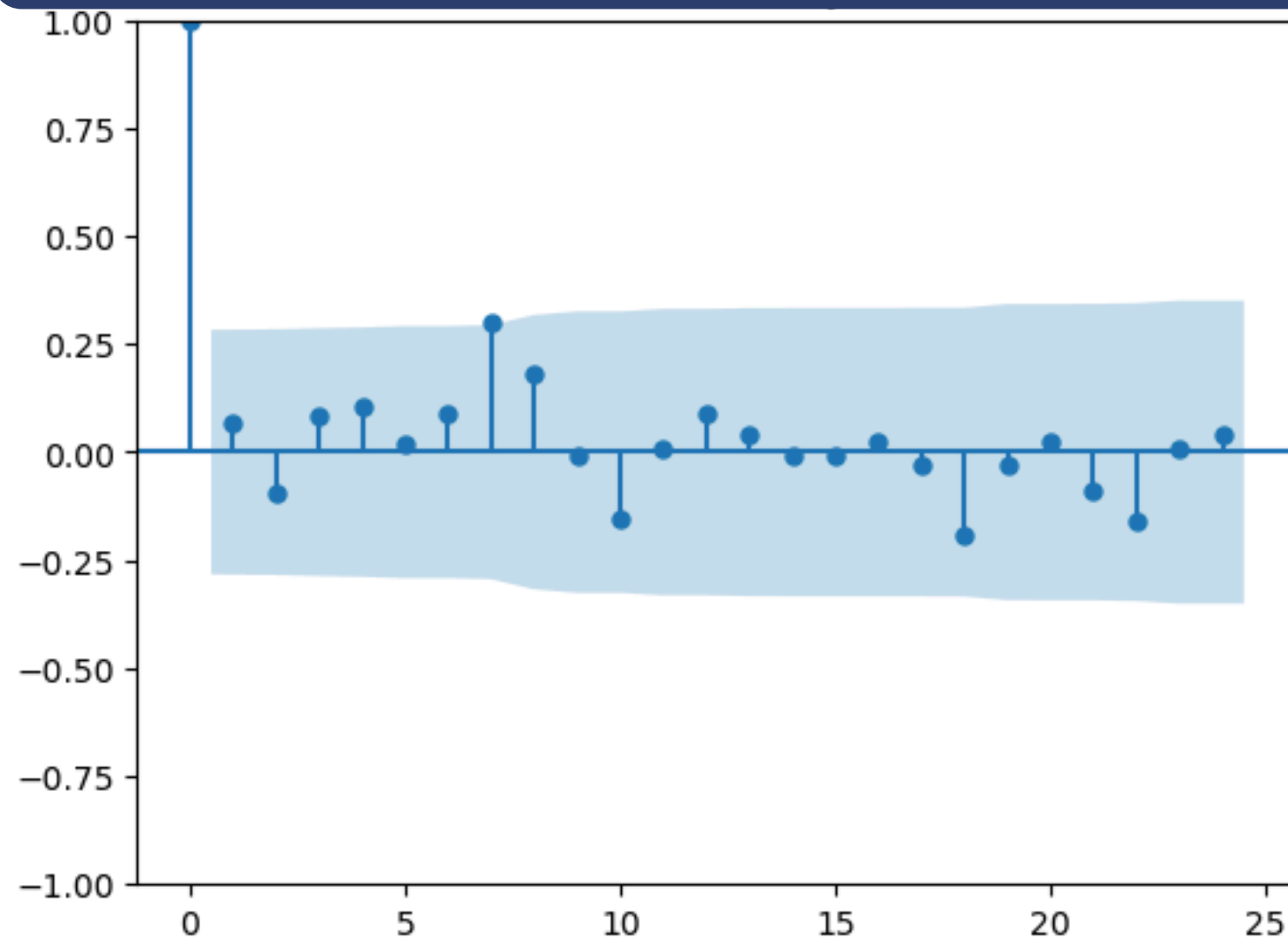
Business Insight:

- Identified an overall upward sales trend
- Detected recurring seasonal sales perfluctuations
- Trend and seasonality strongly influenced sales performance
- Residual variations may reflect external business factors
- Insights support forecasting and business planning

Sales correlation & Forecasting Readiness

Autocorrelation analysis was conducted to evaluate sales dependency across time periods and assess forecasting suitability of the retail sales data.

ACF Plot of Monthly Sales



Key Insight:

- Strong correlation was observed in early lag periods
- Past sales patterns influenced future sales performance
- Correlation gradually declined across higher lags
- Results supported application of time series forecasting models

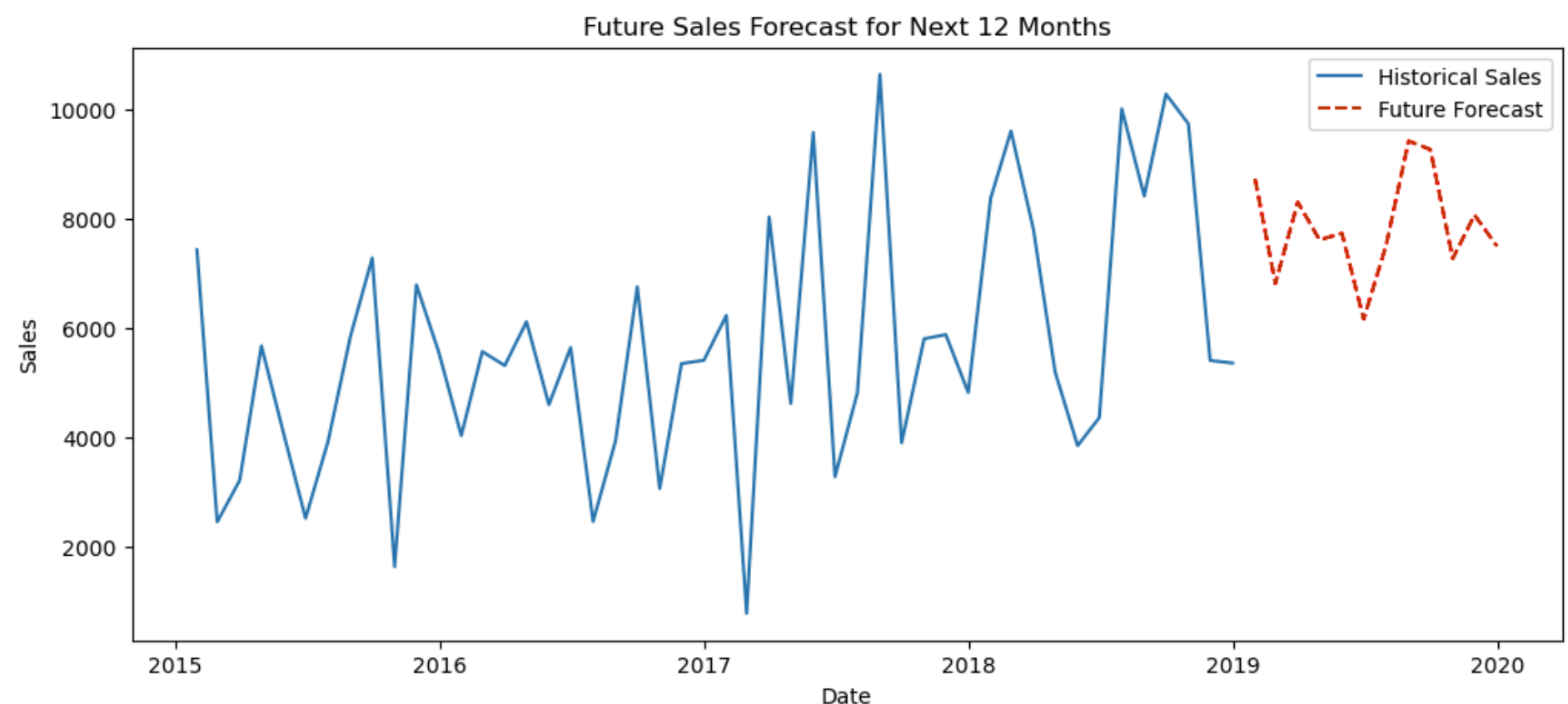
Business Insight:

The observed autocorrelation patterns indicate that historical sales behavior can be effectively utilized for future sales forecasting and demand planning.

12-Month Sales Forecast

A 12-month sales forecast was generated using time series forecasting techniques to predict future retail sales trends.

12- month sales forecast



Forecast Insights :

- Future sales are expected to remain relatively stable over the forecast period
- Forecast results indicate continuing seasonal sales fluctuations
- Predicted sales levels remain consistent with historical sales behavior
- No major declining trend was observed in forecasted sales performance

Business Insight:

Forecasting results can support inventory planning, resource allocation, and strategic business decision-making for future retail operations.

How This Helps Businesses:



Support inventory planing



Improve budgeting and financial planning



Optimize supply chain and resource allocation



Enables data-driven business decisions

Key Insights & Business Recommendation

Key Insights :

- Sales demonstrated a consistent upward trend, indicating sustained business growth and increasing customer demand.
- Distinct seasonal peaks and troughs were observed, highlighting recurring customer purchasing behavior throughout the year.
- Autocorrelation analysis revealed that historical sales significantly influence future sales performance, supporting reliable forecasting.
- ARIMA forecasting predicts stable future demand with recurring seasonal fluctuations, enabling proactive business planning.
- Interactive Power BI dashboards transformed raw sales data into actionable insights for performance monitoring and strategic decision-making.

Business Recommendation:

- ✓ Increase inventory before peak-demand periods.
- ✓ Use forecasting results for procurement and budgeting decisions.
- ✓ Launch targeted marketing campaigns during high-sales seasons.
- ✓ Monitor unusual sales fluctuations to identify potential risks and opportunities.
- ✓ Continuously track sales performance through dashboard reporting.